

# Homework Template

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## Problem 1. Five-card poker hand probabilities

Let the sample space be all 5-card hands from a 52-card deck:

$$\binom{52}{5} = 2,598,960.$$

**(a) Two pairs.** Choose the two ranks for the pairs, then suits for each pair, then the rank and suit of the fifth card (not matching either pair rank).

$$\# = \binom{13}{2} \binom{4}{2} \binom{4}{2} \cdot 11 \cdot 4 = 123,552.$$

$$\Pr(\text{two pairs}) = \frac{123,552}{2,598,960} \approx 0.047539.$$

**(b) Flush (not a straight flush).** Choose a suit, choose 5 of its 13 cards, then subtract straight flushes in that suit.

$$\# = 4 \binom{13}{5} - 40 = 5,108.$$

$$\Pr(\text{flush}) = \frac{5,108}{2,598,960} \approx 0.001965.$$

**(c) Straight (Ace low or high, not a straight flush).** There are 10 possible consecutive rank sets; for each, choose any suits except the 4 same-suit cases.

$$\# = 10(4^5 - 4) = 10,200.$$

$$\Pr(\text{straight}) = \frac{10,200}{2,598,960} \approx 0.003925.$$

**(d) Straight flush.** Ten rank sequences, 4 suits each.

$$\# = 10 \cdot 4 = 40.$$

$$\Pr(\text{straight flush}) = \frac{40}{2,598,960} \approx 0.0000154.$$

**(e) Royal flush.** A royal flush is the highest straight flush; one per suit.

$$\# = 4.$$

$$\Pr(\text{royal flush}) = \frac{4}{2,598,960} \approx 0.00000154.$$

## Problem 2. Roulette (American wheel)

An American roulette wheel has 38 pockets: 18 red, 18 black, and 0, 00 (neither). Each pocket is equally likely, so the sample space size is 38.

1. **Red number.**

$$\Pr(\text{red}) = \frac{18}{38} = \frac{9}{19}.$$

2. **Black twice in a row.** Independence across spins:

$$\Pr(\text{black then black}) = \left(\frac{18}{38}\right)^2 = \left(\frac{9}{19}\right)^2 = \frac{81}{361}.$$

3. **Landing on 0 or 00.**

$$\Pr(\{0, 00\}) = \frac{2}{38} = \frac{1}{19}.$$

4. **In five spins, never landing on 0 or 00.** On any spin, avoid  $\{0, 00\}$  with probability  $36/38 = 18/19$ ; raise to the fifth power:

$$\Pr(\text{no 0 or 00 in 5}) = \left(\frac{18}{19}\right)^5 = \frac{18^5}{19^5} = \frac{1,889,568}{2,476,099}.$$

5. **First spin is one of  $\{1, 2, 3, 4, 5, 6\}$  and next spin is none of them.**

$$\Pr(\text{first} \in \{1, \dots, 6\}) = \frac{6}{38} = \frac{3}{19}, \quad \Pr(\text{next} \notin \{1, \dots, 6\}) = \frac{32}{38} = \frac{16}{19}.$$

By independence,

$$\Pr = \frac{3}{19} \cdot \frac{16}{19} = \frac{48}{361}.$$

## Problem 3. Relative order in a random permutation of $\{1, 2, 3, 4\}$

Let  $\mathcal{S}_4$  be all  $4! = 24$  permutations, each equally likely. For any two distinct symbols  $i, j$ , the events “ $i$  precedes  $j$ ” and “ $j$  precedes  $i$ ” are symmetric, so each has probability  $1/2$ .

1. **1 precedes 4.**

$$\Pr = \frac{1}{2} = \frac{12}{24}.$$

2. **4 precedes 1.**

$$\Pr = \frac{1}{2}.$$

3. **4 precedes 1 and 4 precedes 2.** Consider the relative order among  $\{4, 1, 2\}$ :  $3! = 6$  orders equally likely, and exactly 2 of them have 4 before both 1 and 2 (namely  $4-1-2$  and  $4-2-1$ ).

$$\Pr = \frac{2}{6} = \frac{1}{3}.$$

4. **4 precedes 1, 2, 3 (i.e., 4 is earliest among all four).** By symmetry among the four symbols,

$$\Pr = \frac{1}{4}.$$

5. **4 precedes 3 and 2 precedes 1.** These are two independent pairwise-order constraints, each with probability  $1/2$ :

$$\Pr = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

## Problem 4. Conditional probability of four heads in five flips

We flip a fair coin 5 times. We want:

$$\Pr(\text{exactly } 4H \mid \text{first flip condition}).$$

**(a) Given that the first flip is heads.**

If the first flip is heads, then we need 3 more heads in the remaining 4 flips. Number of ways:  $\binom{4}{3}$ .

Total possible outcomes for the last 4 flips:  $2^4$ .

$$\Pr(\text{exactly } 4H \mid \text{first is H}) = \frac{\binom{4}{3}}{2^4} = \frac{4}{16} = \frac{1}{4}.$$

**(b) Given that the first flip is tails.**

If the first flip is tails, we now need 4 heads in the remaining 4 flips. Number of ways:  $\binom{4}{4} = 1$ .

Total possible outcomes for the last 4 flips:  $2^4$ .

$$\Pr(\text{exactly } 4H \mid \text{first is T}) = \frac{\binom{4}{4}}{2^4} = \frac{1}{16}.$$

## Problem 5. Independence for child-gender events

Assume  $n$  children, each independently boy/girl with probability  $1/2$ .

Let

$$E = \{\text{both sexes occur}\}, \quad F = \{\text{at most one boy}\}.$$

Basic probabilities:

$$\Pr(E) = 1 - \Pr(\text{all same}) = 1 - 2 \cdot 2^{-n} = 1 - 2^{1-n}.$$

Let  $B \sim \text{Bin}(n, \frac{1}{2})$  be the number of boys. Then

$$\Pr(F) = \Pr(B = 0) + \Pr(B = 1) = \frac{\binom{n}{0} + \binom{n}{1}}{2^n} = \frac{1 + n}{2^n}.$$

Moreover,

$$E \cap F = \{\text{both sexes and at most one boy}\} = \{\text{exactly one boy}\},$$

so

$$\Pr(E \cap F) = \Pr(B = 1) = \frac{\binom{n}{1}}{2^n} = \frac{n}{2^n}.$$

$E$  and  $F$  are independent iff

$$\frac{n}{2^n} = \left(1 - 2^{1-n}\right) \frac{n+1}{2^n} \iff 2^{n-1} = n+1.$$

This holds only for  $n = 3$ . For the requested values:

1.  $n = 2$ :  $\Pr(E) = \frac{1}{2}$ ,  $\Pr(F) = \frac{3}{4}$ ,  $\Pr(E \cap F) = \frac{1}{2}$ . Since  $\frac{1}{2} \neq \frac{1}{2} \cdot \frac{3}{4} = \frac{3}{8}$ , **not independent**.
2.  $n = 4$ :  $\Pr(E) = \frac{7}{8}$ ,  $\Pr(F) = \frac{5}{16}$ ,  $\Pr(E \cap F) = \frac{1}{4}$ . Since  $\frac{1}{4} \neq \frac{7}{8} \cdot \frac{5}{16} = \frac{35}{128}$ , **not independent**.
3.  $n = 5$ :  $\Pr(E) = \frac{15}{16}$ ,  $\Pr(F) = \frac{3}{16}$ ,  $\Pr(E \cap F) = \frac{5}{32}$ . Since  $\frac{5}{32} \neq \frac{15}{16} \cdot \frac{3}{16} = \frac{45}{256}$ , **not independent**.

## Problem 6. Bonferroni's inequality for two events

For any two events  $E$  and  $F$ , show that

$$\Pr(E \cap F) \geq \Pr(E) + \Pr(F) - 1.$$

*Proof.* Start from the inclusion–exclusion formula:

$$\Pr(E \cup F) = \Pr(E) + \Pr(F) - \Pr(E \cap F).$$

Since probabilities cannot exceed 1, we have

$$\Pr(E \cup F) \leq 1.$$

Substituting into the first equation gives

$$1 \geq \Pr(E) + \Pr(F) - \Pr(E \cap F).$$

Rearranging terms yields

$$\Pr(E \cap F) \geq \Pr(E) + \Pr(F) - 1,$$

which is Bonferroni's inequality. □

## Problem 7. Generalized Bonferroni inequality

For events  $E_1, \dots, E_n$ ,

$$\Pr\left(\bigcap_{i=1}^n E_i\right) \geq \sum_{i=1}^n \Pr(E_i) - (n-1).$$

*Proof.* We use induction on  $n$ .

*Base case*  $n = 2$ : This is exactly Problem 6.

*Inductive step:* Assume the claim holds for  $n-1 \geq 2$ . Let  $A = \bigcap_{i=1}^{n-1} E_i$ . By the  $n = 2$  case (Bonferroni) applied to  $A$  and  $E_n$ ,

$$\Pr(A \cap E_n) \geq \Pr(A) + \Pr(E_n) - 1.$$

By the induction hypothesis,

$$\Pr(A) = \Pr\left(\bigcap_{i=1}^{n-1} E_i\right) \geq \sum_{i=1}^{n-1} \Pr(E_i) - (n-2).$$

Combining,

$$\Pr\left(\bigcap_{i=1}^n E_i\right) \geq \left(\sum_{i=1}^{n-1} \Pr(E_i) - (n-2)\right) + \Pr(E_n) - 1 = \sum_{i=1}^n \Pr(E_i) - (n-1).$$

Thus the result holds for  $n$ , completing the induction. □

## References